

Design: IPTorrents support (via the Jackett sidecar + FlareSolverr)

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Classification: project-specific

IPTorrents is one of the operator's 4 required trackers and is **not yet supported**. It is a **private English tracker behind Cloudflare**.

1. Why a NATIVE Kotlin provider cannot honestly support IPTorrents today

The Lava tracker SDK's providers use **OkHttp + Jsoup** (plain HTTP client + HTML parser). IPTorrents gates every request behind **Cloudflare's JS/Turnstile challenge**, which a plain HTTP client cannot pass — it requires a real browser engine to execute the challenge JS and obtain the clearance cookie.

Two hard consequences (stated as fact, not guess — §11.4.6):

1. **No reachable HTML to parse.** From an OkHttp client the IPTorrents pages return the Cloudflare interstitial, not the tracker HTML. The dossier (docs/qa/jackett-local-stack-research.md) documents FlareSolverr as the required Cloudflare-bypass for exactly this class of tracker.
2. **Parsers would be guesses.** Writing IPTorrentsSearchParser/... TopicParser without real, reachable fixture HTML means hand-fabricating selectors against HTML we cannot fetch — a **§11.4 PASS-bluff by construction** (tests would assert against invented HTML, proving nothing about the real site). This is forbidden.

Therefore a native provider is **not** the honest path.

2. Recommended path — IPTorrents via the Jackett sidecar + FlareSolverr

Jackett already ships an **IPTorrents Cardigann definition** and drives **FlareSolverr** to clear the Cloudflare challenge. Lava already speaks Jackett's **Torznab** API via lava-api-go (internal/jackett client + GET /jackett/search route, landed 05ecd014). So IPTorrents is reachable through the existing sidecar with **no new HTML parser**:

flowchart LR

```
App["Lava Android app"] -->|"search"| API["lava-api-go<br/>GET /jackett/search?indexer=iptorrents"]
API -->|"Torznab HTTP + apikey"| J["Jackett sidecar<br/>(IPTorrents Cardigann def)"]
J -->|"Cloudflare challenge"| FS["FlareSolverr<br/>(headless browser)"]
FS -->|"clearance cookie"| J
J -->|"authenticated request"| IPT["IPTorrents.com"]
IPT -->|"results HTML"| J
J -->|"Torznab RSS:<br/>enclosure .torrent + magneturl + infohash"| API
API -->|"SearchResult JSON"| App
```

The app sees **one endpoint** (lava-api-go); Jackett + FlareSolverr + the IPTorrents credentials all stay server-side in the local stack.

3. Lava-side work this needs

1. **Jackett-backed provider surface.** A TrackerDescriptor for IPTorrents whose data routes through lava-api-go's Torznab proxy rather than a native HTML client. Two options:
 - (a) **Server-only:** IPTorrents exists only as a Jackett indexer; the app reaches it via the /jackett/search?indexer=iptorrents route — no Lava core/tracker/iptorrents module at all.
 - (b) **Thin descriptor:** a minimal core/tracker/iptorrents descriptor (authType=FORM_LOGIN, capabilities SEARCH+TORRENT_DOWNLOAD+MAGNET_LINK, apiSupported=true) whose TrackerClient delegates search/download/magnet to the Jackett route — so it appears in the provider list and reuses the app's existing search UI. **Recommended** for UX parity.
2. **Credentials (§6.H).** IPTorrents username/password go into **Jackett's gitignored /config volume** (where Jackett stores per-indexer creds), never into Lava source or the app. The operator-provided creds are already staged in the gitignored .env (IPTORRENTS_USERNAME/IPTORRENTS_PASSWORD) for the eventual Jackett-config automation.

3. **FlareSolvrr profile.** Enable the cloudflare compose profile (docker-compose.jackett.yml) so FlareSolvrr runs alongside Jackett (RAM-heavy headless Chromium — off by default, on for Cloudflare trackers).
4. **Per-provider verification (§6.G).** A real-stack test that, with the sidecar running, does `GET /jackett/search?indexer=iptorrents&q=<query>` and asserts a real result carrying a valid .torrent enclosure URL and/or magneturl/infoshash — validated by the same lava.common.torrent validators / the Go Torznab parser. Honest SKIP when the sidecar isn't running (§11.4.3). This is the IPTorrents analogue of the rutor crown-jewel proof.

4. Alternative considered — native provider + embedded FlareSolvrr call from Kotlin

Rejected. It would couple the Android APK (or a per-provider extension) to a running FlareSolvrr service anyway (headless browser can't ship in the APK), is heavier than the sidecar, duplicates what Jackett already does (Cardigann def + FlareSolvrr orchestration), and still cannot run fully on-device. The sidecar path reuses Jackett's maintained IPTorrents definition and keeps the Cloudflare complexity server-side.

5. Recommendation

IPTorrents = Jackett-backed (option 3b: thin descriptor delegating to the /jackett/search route), with FlareSolvrr enabled. It is the only honest, testable path (real Torznab results, real .torrent/magnet, validatable evidence) given Cloudflare — and it generalizes to the other ~500 Jackett indexers for free.

Sources verified

- docs/qa/jackett-local-stack-research.md (FlareSolvrr required for IPTorrents' Cloudflare; GPL-2.0; multi-arch image).
- lava-api-go/internal/jackett/ + internal/handlers/v1/jackett.go + internal/router/router.go
 - internal/config/config.go (the Torznab client + /jackett/search route, landed 05ecd014).
- tools/lava-containers/docker-compose.jackett.yml (jackett + flaresolvrr cloudflare profile).
- core/tracker/api/{TrackerDescriptor,TrackerCapability,AuthType}.kt (the descriptor surface).